

Inaccessibility to 911 and 311 Call Log Data and the Limitations it Creates for AI-Driven Pattern Detection: Case Studies of NYC Floods (2025)



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Abstract:

Urban flood hazards have been increasingly analyzed by AI software trained to pull information from a variety of sources, most of which are news articles, NOAA forecasts, and other external sources. However, despite the good documentation of flooding events and the widespread impacts that these sources cover, AI incorporates little to none of individual flood related calls that are logged in 311 and 911 datasets in their outputs. Unfortunately, most AI-software do not have access to these databases in the first place. Oftentimes, this data is incredibly inaccessible without human aid in decoding, filtering, and feeding the relevant information to them. As a result, when asked to comment on trends in NYC flooding for a specific year (e.g. 2025), their responses are almost entirely based on major news articles. This creates limitations in AI-software's ability to accurately determine instances of flooding in NYC for a specific timeframe, as it will only be able to confidently speak on major floods while ignoring instances that did not make headlines. This leads to inaccuracies in analyzing flood trends across NYC, suggesting them to occur much less frequently than they actually do, in addition to mistaking specific months to be flood prone or not. It is especially important for these trends to be adequately recorded, as they are the factors that influence NYC's emergency response protocols. This is a topic that remains relevant as AI becomes increasingly utilized, and as climate change and global warming continue to impact current flood rates.

Methodology:

We aimed to explore the general limitations AI faces by asking three common AI-software (ChatGPT, PerplexityAI, and Microsoft CoPilot) of their existing knowledge of NYC floods and their patterns for the year of 2025 and reviewed where they pulled their information. We also explored where they pulled their information on two recent major flood events, October 30th and July 14th 2025. Then, we exported recent datasets from NYC OpenData, and analyzed ~7 million 911 incident records, along with ~20 million 311 incident records all from 2025. We used RStudio and NYC OpenData's query feature to filter the datasets and make them readable to the AI software. We narrowed the datasets to show us the calls made on July 14th and October 30th, in addition to any other calls marked as flooding for the year 2025. Data was filtered by date, time, borough, and keywords (e.g., "Water," "Sewer," "Rescue", "Flood", etc.), to locate exactly how many calls were made in relation to flooding events. Then we exported the filtered datasets and once again asked the AIs to identify trends based from what they see in the dataset. Finally, we asked them to apply this new data to their original responses to see if the data would sharpen their response or change it entirely.

Limitations:

- Only worked with 3 of the more popular AI companions, PerplexityAI, ChatGPT, Microsoft Copilot, and there is a tendency for AI's to be biased to always agree with what they are told.
- There are little to no flooding calls logged under NYC's 911 call logs, which is strangely unexpected.
- NYC OpenData datasets are not published immediately, they often takes months to a year to be exportable, meaning that AI would be justified in not using them for present-day analysis.

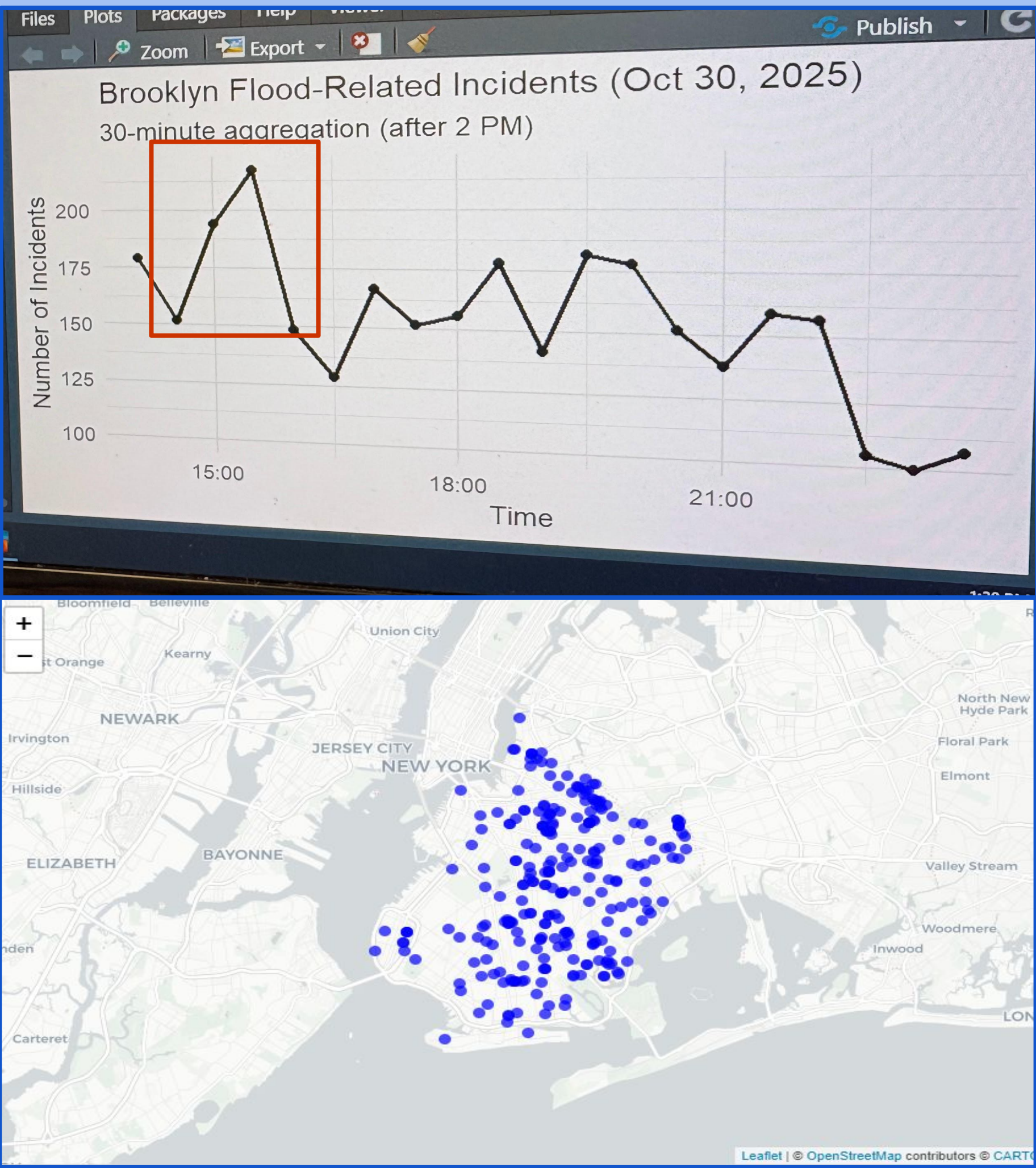


Figure 1: October 30th. Timeline of 911 calls pulled from the 911 call dataset (upper). Map created with Rstudio plotting 311 calls retrieved from a 20 million calls dataset during the flood event in Brooklyn (lower).

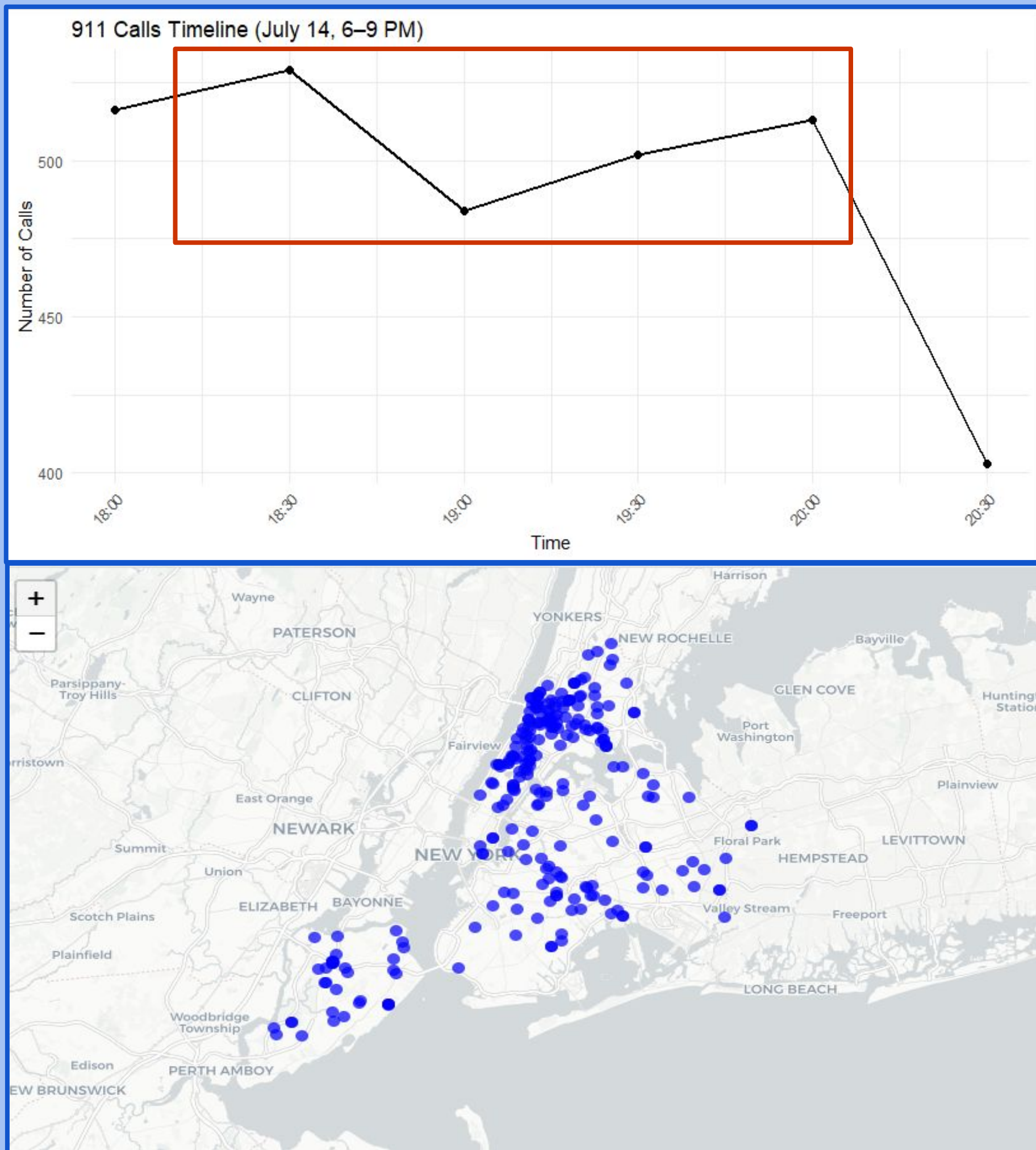


Figure 2: July 14th. Timeline of 911 calls pulled from the 911 call dataset (upper). Map created with Rstudio plotting 311 calls retrieved from 7 million records data set during the flood event in all boroughs (lower).

AI's Outputs Before/After Dataset Knowledge

ChatGPT

- Most recent knowledge of NYC flooding was October 30 and July 14.
- Says flooding is caused by sea level rise, intense rainfall, coastal flooding, and flash floods are the biggest threat.
- Information is pulled from news sources such as NYT, ABC, CBS, NOAA forecasts, YouTube videos, and other articles.
- Does not use 311/911 logs due to inaccessibility, messiness, and delays in availability to public.

- Worked with the dataset well, was able to analyze the data and note additional trends when it was filtered and fed to them manually.
- Removed its generalization of coastal flood domination, as the dataset revealed it was more sewer flood dominated.
- May - August has elevated activity, October dominated the whole year, corrected its underestimation of fall storms.
- Queens and Brooklyn are the epicenter of the flooding.
- Baseline flooding is still present even in quieter months such as December.
- Areas that are low lying in elevation are the first to flood and the last to be addressed.

Perplexity

- Most recent knowledge of NYC flooding was October 30 and July 31.
- Floods caused by intense and frequent rainfall, flashflooding, and drainage incapacity.
- Information pulled from news sources such as The NYT, NOAA reports, FOX weather, or blogs.
- Does not use 311/911 logs due to restriction issues to raw data, only has access to public summaries of them.

- Worked with the dataset well after it was filtered and fed to them manually, pointed out new patterns for the year.
- Noted that most flooding is tied to specific weather events, and drainage hotspots.
- There is a cluster of flooding around mid-late december, second largest spike to October, and that there is a mid-late summer period of flash flooding.
- Queens, Brooklyn, and Staten Island are the most have the most vulnerable neighborhoods based on the complaints.
- Adjusted its generalization of only street flood occurrences to account for water intruding into homes just as frequently.
- Chronic and abundant hot spots of complaints come from wealthier neighborhoods.

Microsoft CoPilot

- Most recent knowledge of NYC flooding was July 14 and October 30.
- Flooding is caused by localized rainfall intensity, lack of drainage capacity, and high tide.
- Information is pulled from Previsico, FOX weather, NY Sea Grant, and other post flooding articles.
- Does not use 311/911 logs due to public restrictions, cannot see the data unless it is cited in news sources, and the size of unfiltered datasets.

- Had some difficulty analyzing the dataset due to its large size after filtering, but when fed the whole dataset split into small files it also found new trends to analyze.
- Confirmed that flooding is rain-fall driven, and flashflooding dominates a large portion of the flooding in NYC.
- Indoor flooding is prominent in the Bronx and Manhattan, while street flooding targets Queens and Brooklyn.
- Spikes occur in October, mid-late summer, and May.
- Removed its assumption that flooding was coastal or high tide related.
- Drainage issues continue to contribute to NYC flooding but mostly in specific corridors.

Results + Discussion

After working with the raw datasets, we found that calls related to flooding are not present within 911 databases. This was unexpected, as we assumed that most people would call 911 to make a report first, especially given that flooding leads to dire situations where help is necessary. When checking for the number of calls on major floods such as Oct 30th and July 14th, instances where people died due to the flooding, we found no marked calls for the incidents. Despite this lack of information, running the entire dataset through Rstudio shows spikes in 911 calls being made at times that are consistent with the times of peak flooding on October 30th and July 14th. This indicates that calls were being made, but they were not recorded adequately within the 911 database. After further analysis, we found that most flood related calls/complaints are logged under 311 databases instead. We prioritized using this dataset as our main source to feed to the AIs. After feeding the filtered data found in the 311 dataset to the AI-Software, they were more confident in identifying trends, patterns, and instances of flooding in 2025 now that they had real-time data to cross reference with their original sources. All of them added information to their initial responses that they could not have seen without access to the dataset, which proves that inaccessibility is a real limitation.

As seen, all three AI software pull their information from news sources, videos, NOAA forecasts/reports, and other general articles, which explains why their most recent knowledge of flooding in NYC were the two biggest instances of the whole year. Giving them access to the dataset allowed them to expand their knowledge and pick up on flooding in other months that did not make headlines, allowing them to correct any media bias and underrepresentation that influenced their previously limited understanding. They were able to look into specific details surrounding 2025 flooding that are not generally available.

Additionally, the AI's not only pulled new information about the seasonality of flooding, but also which intersections were the most vulnerable. Most of the highly vulnerable streets lie within Queens and Brooklyn, with Staten Island coming in third. The Bronx and Manhattan were found to be more associated with indoor flooding.

Conclusion

Overall, AI software display great capability of working with 311 and 911 datasets for analytical purposes if only given the access and filtering. Their results showcase the limitation of inaccessibility that arises for current AI analysis, and leads to the question of how often it presents itself on a bigger scale. Having this data at their disposal allowed AI to correct vague interpretations and expose meaningful connections drawn from flooding trends of 2025. It emphasizes the importance of integrating multi-source data and improving the accessibility of these datasets to AI-Software responsible for making decisions about flooding in NYC. It would allow emergency response decision makers to avoid inaccurate AI-driven decisions and ensure the most accurate information is being used in pattern detection.

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